# AI-POWERED VEHICLE DAMAGE DETECTION

## *Insurance Claims*

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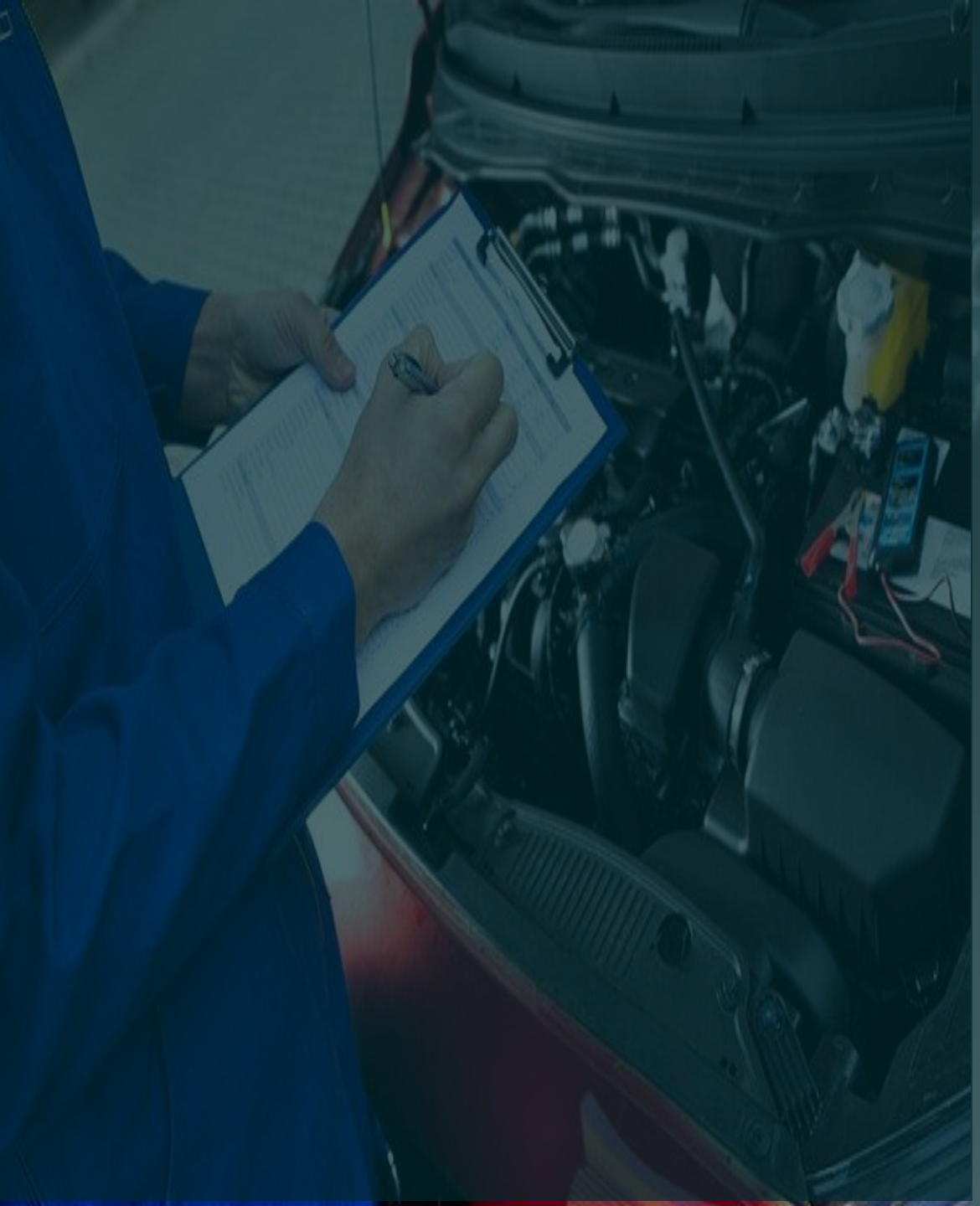
***ITAI 1378 – Computer Vision and AI***
***GitHub Midterm***



# PROBLEM:

- Insurance companies rely on manual inspections for vehicle damage.
- This process is slow, costly, and inconsistent.
- Customers wait days for repair estimates.

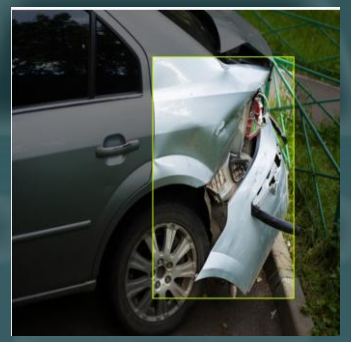
# SOLUTION:

- Users upload a photo of a damaged vehicle.
- AI model detects damage areas (dents, scratches, cracks).
- System classifies the type of damage.
- Generates an estimated repair cost automatically.

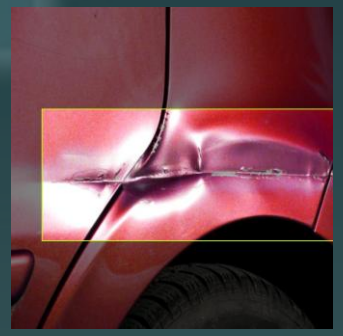
# TECHNICAL APPROACH:

- Technique: Object Detection
- Model: YOLOv8
- Frameworks:
  - PyTorch
  - Ultralytics YOLO
  - OpenCV
- Justification:
  YOLOv8 is fast and accurate, ideal for real-time damage detection.

# DATA PLAN:



- **Dataset:**
  - Roboflow Universe – Automobile Damage Detection
  - ~6,900 labeled images
  - Classes: dent, scratch, crack



- **Preparation:**
  - Export in YOLO format
  - Split into train/validation/test
  - Optional data augmentation

# PIPELINE:

Vehicle Image

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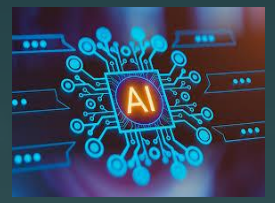


Image Preprocessing

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Damage Detection (YOLOv8)

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Damage Classification

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Repair Cost Estimation

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Output Report

# METRICS:

**Primary Metric:**

- Detection accuracy ≥ 85%

**Secondary Metrics:**

- Damage classification accuracy ≥ 80%
- Inference time < 1 second per image

# WEEK-BY-WEEK PLAN:

| Week | Task | Milestone |
|---|---|---|
| Week 10 | Dataset collection | Dataset ready |
| Week 11 | Train object detection model | Model running |
| Week 12 | Improve accuracy | Stable performance |
| Week 13 | Add cost estimation module | Demo ready |
| Week 14 | Testing & documentation | Final version |
| Week 15 | Presentation | Project demonstration |

# CHALLENGES & BACKUP PLANS:

| Challenge | Plan B |
|---|---|
| Dataset too small | Use alternative Roboflow datasets |
| Low model accuracy | Apply data augmentation or fine-tune larger YOLO model |
| Training too slow | Use Google Colab GPU |

# RESOURCES NEEDED:

- Compute: Google Colab GPU
- Frameworks:
  - PyTorch
  - Ultralytics YOLO
  - OpenCV
- Estimated Cost: $0
- Optional Demo: Streamlit web app